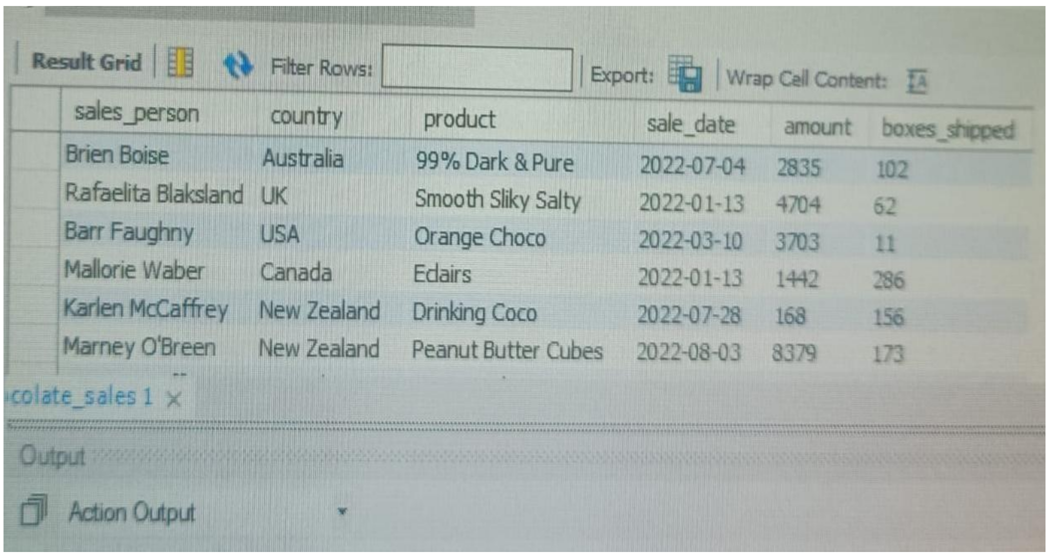
SQL and Excel Assignment

Task 1: Introduction to Databases & SELECT Statement

1a. Display all records

```
SELECT *
FROM chocolate_sales;
```

Output



sales_person	country	product	sale_date	amount	boxes_shipped
Brien Boise	Australia	99% Dark & Pure	2022-07-04	2835	102
Rafaelita Blaksland	UK	Smooth Sliky Salty	2022-01-13	4704	62
Barr Faughny	USA	Orange Choco	2022-03-10	3703	11
Mallorie Waber	Canada	Eclairs	2022-01-13	1442	286
Karlen McCaffrey	New Zealand	Drinking Coco	2022-07-28	168	156
Marney O'Breen	New Zealand	Peanut Butter Cubes	2022-08-03	8379	173

- This query retrieves every column and every row from the chocolate_sales table.
- The asterisk (*) is a wildcard that tells SQL to return all available columns without listing them individually.
- It is typically used early on to get a quick overview of the full dataset.
- In large tables this should be used carefully, since it can return a very high volume of data.

1b. Select specific columns

```
SELECT sales_person, country, product, amount
FROM chocolate_sales;
```

Result Grid | Filter Rows: | Export: | Wrap Ce

sales_person	country	product	amount
Jehu Rudeforth	UK	Mint Chip Choco	5320
Van Tuxwell	India	85% Dark Bars	7896
Gigi Bohling	India	Peanut Butter Cubes	4501
Jan Morforth	Australia	Peanut Butter Cubes	12726
Jehu Rudeforth	UK	Peanut Butter Cubes	13685
Van Tuxwell	India	Smooth Sliky Salty	5376
Oby Sorrel	UK	99% Dark & Pure	13685
Gunar Cockshoot	Australia	After Nines	3080
Jehu Rudeforth	New Zealand	50% Dark Bites	3990
Brien Boise	Australia	99% Dark & Pure	2835

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- This query selects only the four named columns instead of the entire table.
- Selecting specific columns reduces the amount of data returned and makes results easier to read.
- It is good practice once you know exactly which fields are relevant to your analysis.
- The column order in the SELECT list determines the order in which they appear in the output.

1c. Select specific columns using column aliases

```
SELECT
    sales_person  AS "Sales Rep",
    country       AS "Country",
    product       AS "Product Name",
    amount        AS "Sale Amount ($)"
FROM chocolate_sales;
```

Output:

Export: | Wrap Cell

Sales Rep	Country	Product Name	Sale Amount ($)
Jehu Rudeforth	UK	Mint Chip Choco	5320
Van Tuxwell	India	85% Dark Bars	7896
Gigi Bohling	India	Peanut Butter Cubes	4501
Jan Morforth	Australia	Peanut Butter Cubes	12726
Jehu Rudeforth	UK	Peanut Butter Cubes	13685
Van Tuxwell	India	Smooth Sliky Salty	5376
Oby Sorrel	UK	99% Dark & Pure	13685
Gunar Cockshoot	Australia	After Nines	3080
Jehu Rudeforth	New Zealand	50% Dark Bites	3990

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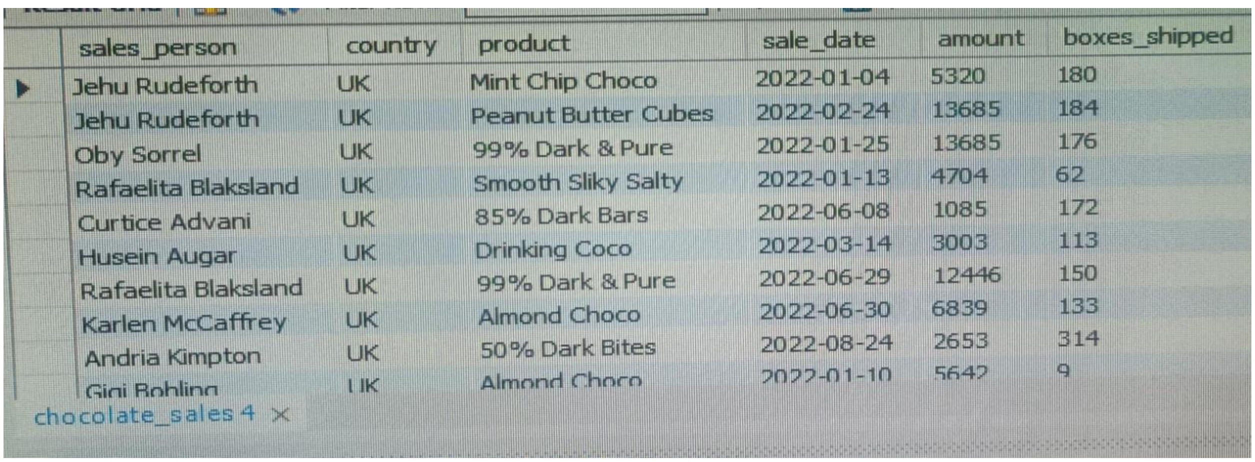
- The AS keyword is used to give a column a temporary, more descriptive name in the result set.
- Aliases do not change the underlying table structure; they only affect how the output is labeled.
- This is useful for making reports more readable, especially for non-technical audiences.
- Aliases containing spaces are wrapped in double quotes so SQL treats them as a single identifier.

Task 2: WHERE, ORDER BY & Aggregate Functions

2a. Filter records using the WHERE clause

```
SELECT *
FROM chocolate_sales
WHERE country = 'UK';
```

Output



sales_person	country	product	sale_date	amount	boxes_shipped
Jehu Rudeforth	UK	Mint Chip Choco	2022-01-04	5320	180
Jehu Rudeforth	UK	Peanut Butter Cubes	2022-02-24	13685	184
Oby Sorrel	UK	99% Dark & Pure	2022-01-25	13685	176
Rafaelita Blaksland	UK	Smooth Sliky Salty	2022-01-13	4704	62
Curtice Advani	UK	85% Dark Bars	2022-06-08	1085	172
Husein Augar	UK	Drinking Coco	2022-03-14	3003	113
Rafaelita Blaksland	UK	99% Dark & Pure	2022-06-29	12446	150
Karlen McCaffrey	UK	Almond Choco	2022-06-30	6839	133
Andria Kimpton	UK	50% Dark Bites	2022-08-24	2653	314
Gigi Bohling	UK	Almond Choco	2022-01-10	5642	9

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- The WHERE clause filters rows so only those matching the given condition are returned.
- Here, only sales made in the UK are included in the result set.
- WHERE is evaluated before grouping or sorting, so it works on raw table rows.
- Text values such as 'UK' must be enclosed in single quotes in standard SQL.

2b. Use comparison operators in conditions

```
SELECT sales_person, product, amount
FROM chocolate_sales
WHERE amount > 10000;
```

Output

sales_person	product	amount
Jan Morforth	Peanut Butter Cubes	12726
Jehu Rudeforth	Peanut Butter Cubes	13685
Oby Sorrel	99% Dark & Pure	13685
Rafaelita Blaksland	99% Dark & Pure	12446
Karlen McCaffrey	Raspberry Choco	14749
Beverie Moffet	Raspberry Choco	14798
Brien Boise	99% Dark & Pure	16793
Barr Faughny	Fruit & Nut Bars	15421
Oby Sorrel	99% Dark & Pure	10255
Van Tuxwell	Organic Choco Syrup	19453

- Comparison operators like >, <, =, >=, and <= let you filter numeric or date values.
- This query returns only transactions where the sale amount exceeds 10,000.
- Comparison operators can be combined using AND / OR for more complex conditions.
- They are essential for isolating outliers, top performers, or records above a threshold.

2c. Sort records using ORDER BY

```
SELECT sales_person, product, amount
FROM chocolate_sales
ORDER BY amount DESC;
```

Output

sales_person	product	amount
Ches Bonnell	Peanut Butter Cubes	26171
Ches Bonnell	Peanut Butter Cubes	23252
Van Tuxwell	Organic Choco Syrup	22943
Van Tuxwell	Organic Choco Syrup	22751
Curtice Advani	Smooth Sliky Salty	22717
Van Tuxwell	Organic Choco Syrup	22702
Marney O'Breen	Smooth Sliky Salty	22559
Ches Bonnell	Peanut Butter Cubes	22050
Kaine Padly	After Nines	21813
Van Tuxwell	Organic Choco Syrup	21763

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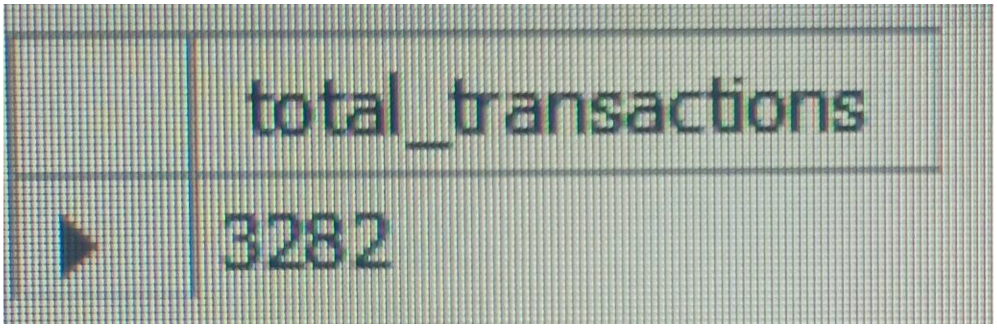
- ORDER BY sorts the result set based on one or more columns.
- DESC sorts from highest to lowest; ASC (the default) sorts from lowest to highest.
- Sorting happens after filtering, so WHERE conditions are applied first.
- This query highlights the largest sales transactions at the top of the output.

2d. Aggregate function: COUNT()

```
SELECT COUNT(*) AS total_transactions
FROM chocolate_sales;
```

Output



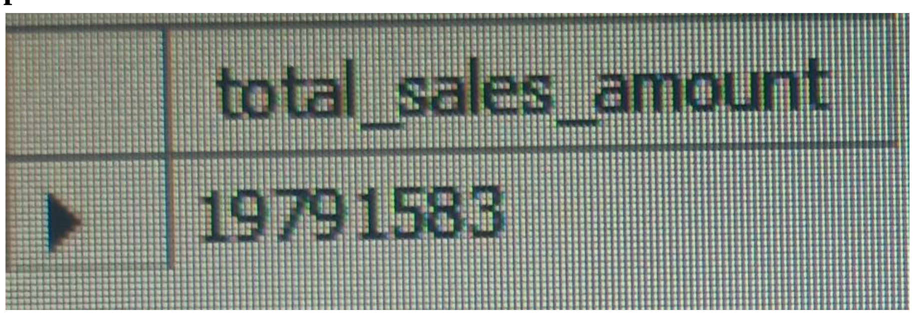
total_transactions
3282

- COUNT(*) returns the total number of rows in the table or query result.
- It is commonly used to measure the size of a dataset or a filtered subset of it.
- COUNT can also be applied to a specific column to count only non-NULL values.
- Here it tells us how many total sales transactions exist in the dataset.

2e. Aggregate function: SUM()

```
SELECT SUM(amount) AS total_sales_amount
FROM chocolate_sales;
```

Output



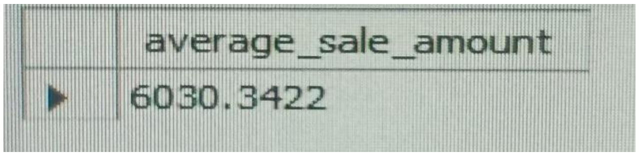
total_sales_amount
19791583

- SUM() adds up all numeric values in the specified column.
- This query calculates the total revenue generated across all recorded sales.
- SUM ignores NULL values automatically when performing the calculation.
- It is one of the most common aggregate functions used in financial reporting.

2f. Aggregate function: AVG()

```
SELECT AVG(amount) AS average_sale_amount
FROM chocolate_sales;
```

Output:



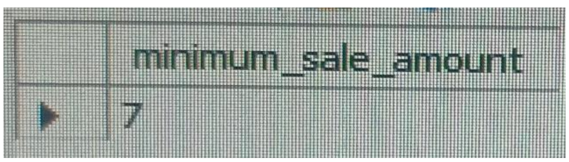
average_sale_amount
6030.3422

- AVG() calculates the mean value of a numeric column across all rows.
- This query shows the average amount generated per sales transaction.
- Like SUM, AVG ignores NULL values when computing the result.
- Average values are useful for understanding typical transaction size.

2g. Aggregate function: MIN()

```
SELECT MIN(amount) AS minimum_sale_amount
FROM chocolate_sales;
```

Output



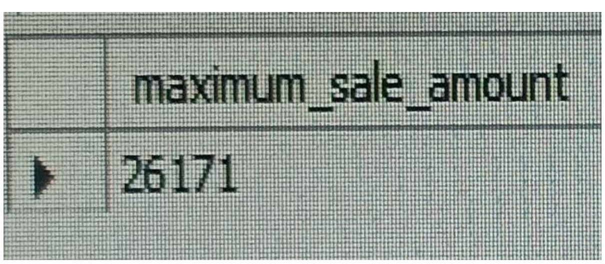
minimum_sale_amount
7

- MIN() returns the smallest value found in the specified column.
- This query identifies the lowest single sale amount in the dataset.
- MIN can be applied to numeric, date, or text columns.
- It is often used to spot unusually low or possibly erroneous entries.

2h. Aggregate function: MAX()

```
SELECT MAX(amount) AS maximum_sale_amount
FROM chocolate_sales;
```

Output



maximum_sale_amount
26171

- MAX() returns the largest value found in the specified column.
- This query identifies the single highest sale amount in the dataset.
- MAX is frequently paired with MIN to describe the full range of values.
- It helps highlight top-performing transactions or products.

Task 3: GROUP BY & HAVING

3a. Group records using GROUP BY

```
SELECT
    country,
    COUNT(*)      AS total_orders,
    SUM(amount)   AS total_sales
FROM chocolate_sales
GROUP BY country
ORDER BY total_sales DESC;
```

Output

country	total_orders	total_sales
Australia	615	3646458
UK	534	3365384
India	552	3343734
USA	537	3313861
Canada	525	3078504
New Zealand	519	3043642

- GROUP BY collapses multiple rows sharing the same country into a single summary row.
- Aggregate functions like COUNT and SUM are then calculated once per group rather than once per row.
- This query shows the number of orders and total sales for each country.
- GROUP BY must include every non-aggregated column that appears in the SELECT list.

3b. Use HAVING to filter grouped results

```
SELECT
    country,
    SUM(amount) AS total_sales
FROM chocolate_sales
GROUP BY country
HAVING SUM(amount) > 1000000
ORDER BY total_sales DESC;
```

Output

country	total_sales
Australia	3646458
UK	3365384
India	3343734
USA	3313861
Canada	3078504
New Zealand	3043642

- HAVING filters groups after aggregation, whereas WHERE filters rows before aggregation.
- This query keeps only countries whose combined total sales exceed 1,000,000.
- HAVING conditions typically reference an aggregate function, such as SUM or COUNT.
- It is the correct clause to use whenever a filter depends on a group-level calculation.

Task 4: SQL Joins

4a. INNER JOIN

```
SELECT
    cs.sales_person,
    cs.product,
    pc.category,
    cs.amount
FROM chocolate_sales cs
INNER JOIN product_category pc
    ON cs.product = pc.product;
```

Output

sales_person	product	category	amount
Jehu Rudeforth	Mint Chip Choco	Chocolate	5320
Van Tuxwell	85% Dark Bars	Dark Chocolate	7896
Gigi Bohling	Peanut Butter Cubes	Snack	4501
Jan Morforth	Peanut Butter Cubes	Snack	12726
Jehu Rudeforth	Peanut Butter Cubes	Snack	13685
Van Tuxwell	Smooth Sliky Salty	Chocolate	5376
Oby Sorrel	99% Dark & Pure	Dark Chocolate	13685
Gunar Cockshoot	After Nines	Chocolate	3080
Jehu Rudeforth	50% Dark Bites	Dark Chocolate	3990
Brien Boise	99% Dark & Pure	Dark Chocolate	2835

- INNER JOIN returns only the rows where the join column matches in both tables.
- Here, only sales whose product has a matching entry in product_category are returned.
- Rows from either table with no match on the other side are excluded from the result.

- INNER JOIN is the most commonly used join when you need fully matched data from both tables.

4b. LEFT JOIN

```
SELECT
    cs.sales_person,
    cs.product,
    pc.category,
    cs.amount
FROM chocolate_sales cs
LEFT JOIN product_category pc
    ON cs.product = pc.product;
```

Output

sales_person	product	category	amount
Jehu Rudeforth	Mint Chip Choco	Chocolate	5320
Van Tuxwell	85% Dark Bars	Dark Chocolate	7896
Gigi Bohling	Peanut Butter Cubes	Snack	4501
Jan Morforth	Peanut Butter Cubes	Snack	12726
Jehu Rudeforth	Peanut Butter Cubes	Snack	13685
Van Tuxwell	Smooth Sliky Salty	Chocolate	5376
Oby Sorrel	99% Dark & Pure	Dark Chocolate	13685
Gunar Cockshoot	After Nines	Chocolate	3080
Jehu Rudeforth	50% Dark Bites	Dark Chocolate	3990
Brien Boise	99% Dark & Pure	Dark Chocolate	2835

- LEFT JOIN returns all rows from the left table (chocolate_sales), regardless of a match.
- Where no matching product_category row exists, the category column is returned as NULL.
- This is useful for auditing sales records that are missing a product classification.
- The left table is always fully preserved, unlike INNER JOIN which can drop unmatched rows.

4c. RIGHT JOIN

```
SELECT
    pc.product,
    pc.category,
    cs.sales_person,
    cs.amount
FROM chocolate_sales cs
RIGHT JOIN product_category pc
    ON cs.product = pc.product;
```

Output

product	category	sales_person	amount
50% Dark Bites	Dark Chocolate	Roddy Speechley	9976
50% Dark Bites	Dark Chocolate	Beverie Moffet	2922
50% Dark Bites	Dark Chocolate	Rafaelita Blaksland	11491
50% Dark Bites	Dark Chocolate	Curtice Advani	11226
50% Dark Bites	Dark Chocolate	Gigi Bohling	6723
50% Dark Bites	Dark Chocolate	Van Tuxwell	4332
50% Dark Bites	Dark Chocolate	Ches Bonnell	12584
50% Dark Bites	Dark Chocolate	Kaine Padly	3402
50% Dark Bites	Dark Chocolate	Oby Sorrel	7302
50% Dark Bites	Dark Chocolate	Oby Sorrel	814

- RIGHT JOIN returns all rows from the right table (product_category), regardless of a match.
- Products that were never sold still appear in the result, with NULL sales-related columns.
- This is useful for identifying catalog items that have not generated any sales.
- RIGHT JOIN behaves like a mirrored LEFT JOIN, with priority given to the right-hand table.

Task 5: SQL Subqueries

5a. Sales persons earning above the average total

```
SELECT sales_person, total_amount
FROM (
    SELECT sales_person, SUM(amount) AS total_amount
    FROM chocolate_sales
    GROUP BY sales_person
) AS person_totals
WHERE total_amount > (
    SELECT AVG(total_amount)
    FROM (
        SELECT SUM(amount) AS total_amount
        FROM chocolate_sales
        GROUP BY sales_person
    ) AS avg_calc
)
ORDER BY total_amount DESC;
```

Output

sales_person	total_amount
Ches Bonnell	1022601
Oby Sorrel	1017207
Madelene Upcott	1010032
Kelci Walkden	1002925
Brien Boise	997327
Van Tuxwell	974421
Dennison Crosswaite	931851
Beverie Moffet	892419
Kaine Padly	849068
Marney O'Breen	836431

- A subquery is a query nested inside another query, used here to first total sales per person.
- The outer query then compares each person's total against the overall average of those totals.
- This two-step approach lets us benchmark individuals against the group average dynamically.
- Subqueries in the FROM clause act like temporary, on-the-fly tables for the outer query to use.

5b. Products with above-average sale amount

```
SELECT product, AVG(amount) AS avg_product_amount
FROM chocolate_sales
GROUP BY product
HAVING AVG(amount) > (
    SELECT AVG(amount)
    FROM chocolate_sales
)
ORDER BY avg_product_amount DESC;
```

Output

product	avg_product_amount
Peanut Butter Cubes	7051.6463
Mint Chip Choco	6703.6667
Choco Coated Almonds	6607.3761
Manuka Honey Choco	6557.1407
99% Dark & Pure	6530.8912
Baker's Choco Chips	6496.5854
85% Dark Bars	6368.4400
Smooth Sliky Salty	6328.8136
Almond Choco	6183.6875
Organic Choco Syrup	6059.9231

- The inner subquery calculates a single value: the overall average sale amount across all rows.
- The outer query groups sales by product and computes each product's own average amount.
- HAVING then keeps only products whose average exceeds the overall average from the subquery.
- This pattern is a common way to identify above-average performers within a larger dataset.

Task 6: Data Formatting, Sorting & Filtering

6a. Format headers and data appropriately

```
Steps performed:
1. Select the header row and apply Bold + fill color.
2. Apply borders to the full data range.
3. Format the Amount column as Currency / Number with thousand separators.
4. Format the Date column using a consistent Date format (dd-mmm-yy).
```

- Proper formatting makes a dataset easier to read and interpret at a glance.
- Bold, colored headers visually separate column labels from the actual data.
- Number and date formatting ensures values are displayed consistently across the sheet.
- Good formatting is also a first step toward avoiding data-entry and calculation errors.

6b. Sort the dataset

```
Steps performed:
1. Select the full data range (including headers).
2. Go to Data tab -> Sort.
3. Sort by Amount, Largest to Smallest (or by Country, A to Z).
4. Confirm 'My data has headers' is checked before sorting.
```

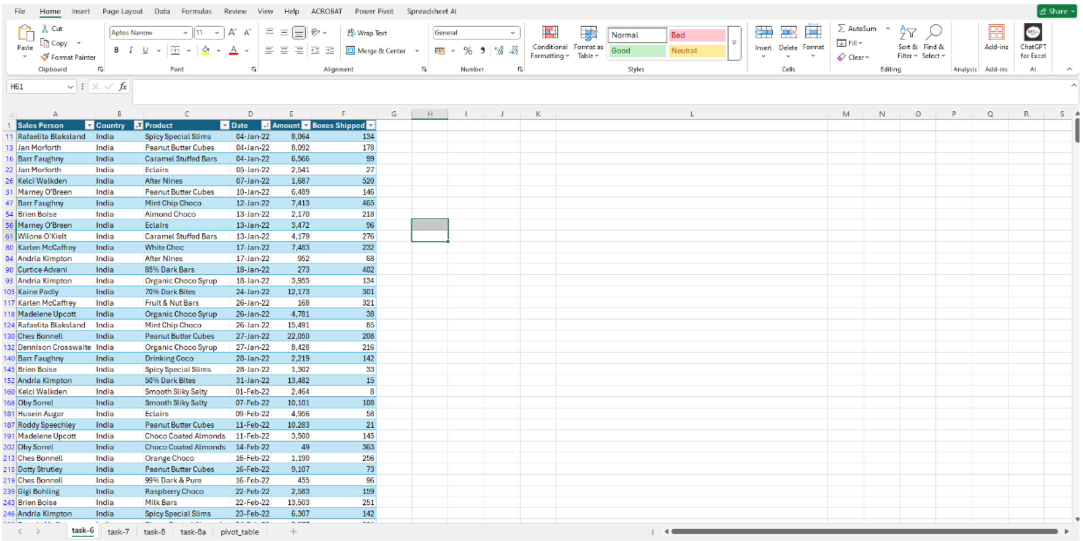
- Sorting reorders rows based on the values in one or more chosen columns.
- It makes it easy to quickly find the highest or lowest values in a dataset.
- Multi-level sorting can be used to sort by one column and then break ties with another.
- Sorting does not delete or alter data — it only changes the row order for display.

6c. Apply filters to filter records based on specific conditions

```
Steps performed:
1. Select the header row and click Data -> Filter.
2. Click the drop-down arrow on the Country column.
3. Choose a specific country (e.g., UK) to display only matching rows.
4. Optionally combine with a Number Filter on Amount (e.g., greater than 10000).
```

- Filtering temporarily hides rows that do not meet the chosen criteria, without deleting them.
- It allows quick, interactive exploration of subsets of a large dataset.
- Multiple filters can be applied across different columns at the same time.
- Filters can be cleared at any point to instantly restore the full dataset view.

Output:



Task 7: Conditional Formatting & Excel Functions

7a. Conditional Formatting — highlight values based on conditions

```
Home tab -> Conditional Formatting -> Highlight Cell Rules
-> Greater Than -> 18000 -> Green Fill

Applied to range: Amount column (e.g., E2:E3283)
```

- Conditional formatting automatically changes a cell's appearance based on its value.
- Here, sale amounts above 10,000 are highlighted so high-value transactions stand out visually.
- The formatting updates automatically if the underlying data changes.
- It is a quick way to spot patterns or outliers without writing a formula.

7b. IF() function

```
=IF(F2>10000,"High","Low")
```

- IF() tests a logical condition and returns one value if TRUE and another if FALSE.
- Here, each sale is labeled 'High' if the amount exceeds 10,000, otherwise 'Low'.
- IF() is one of the most widely used functions for creating simple business rules in Excel.
- Multiple IF() functions can be nested to handle more than two possible outcomes.

7c. COUNTIF() function

```
=COUNTIF(A2:F3283,"canada")
```

- COUNTIF() counts the number of cells in a range that meet a single condition.
- This formula counts how many sales transactions were recorded for the UK.
- It is useful for quickly summarizing how often a category or value occurs in a dataset.

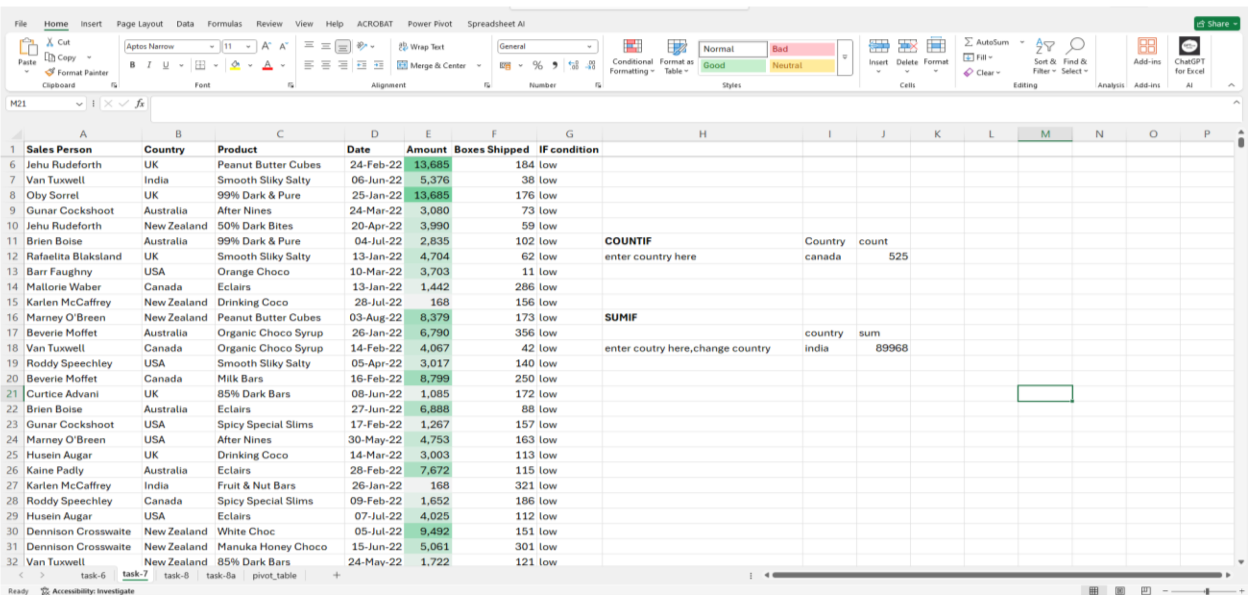
- The condition can use text, numbers, or comparison operators such as ">10000".

7d. SUMIF() function

```
=SUMIF(A2:F3283,"UK",E2:E3283)
```

- SUMIF() adds up values in one range based on a condition applied to another range.
- This formula totals the Amount column, but only for rows where Country equals 'UK'.
- It combines filtering and totaling in a single formula, without needing a helper table.
- SUMIF() is commonly used for category-wise totals such as sales by country or product.

Task-7 output:



1	Sales Person	Country	Product	Date	Amount	Boxes Shipped	IF condition			
6	Jehu Rudeforth	UK	Peanut Butter Cubes	24-Feb-22	13,685	184	low			
7	Van Tuxwell	India	Smooth Sliky Salty	06-Jun-22	5,376	38	low			
8	Oby Sorrel	UK	99% Dark & Pure	25-Jan-22	13,685	176	low			
9	Gunar Cockshoot	Australia	After Nines	24-Mar-22	3,080	73	low			
10	Jehu Rudeforth	New Zealand	50% Dark Bites	20-Apr-22	3,990	59	low			
11	Brien Boise	Australia	99% Dark & Pure	04-Jul-22	2,835	102	low	COUNTIF	Country	count
12	Rafaelita Blaksland	UK	Smooth Sliky Salty	13-Jan-22	4,704	62	low	enter country here	canada	525
13	Barr Faughny	USA	Orange Choco	10-Mar-22	3,703	11	low			
14	Mallorie Waber	Canada	Eclairs	13-Jan-22	1,442	286	low			
15	Karlen McCaffrey	New Zealand	Drinking Coco	28-Jul-22	168	156	low			
16	Marney O'Breen	New Zealand	Peanut Butter Cubes	03-Aug-22	8,379	173	low	SUMIF		
17	Beverie Moffet	Australia	Organic Choco Syrup	26-Jan-22	6,790	356	low		country	sum
18	Van Tuxwell	Canada	Organic Choco Syrup	14-Feb-22	4,067	42	low	enter coutry here,change country	india	89968
19	Roddy Speechley	USA	Smooth Sliky Salty	05-Apr-22	3,017	140	low			
20	Beverie Moffet	Canada	Milk Bars	16-Feb-22	8,799	250	low			
21	Curtice Advani	UK	85% Dark Bars	08-Jun-22	1,085	172	low			
22	Brien Boise	Australia	Eclairs	27-Jun-22	6,888	88	low			
23	Gunar Cockshoot	USA	Spicy Special Slims	17-Feb-22	1,267	157	low			
24	Marney O'Breen	USA	After Nines	30-May-22	4,753	163	low			
25	Husein Augar	UK	Drinking Coco	14-Mar-22	3,003	113	low			
26	Kaine Padly	Australia	Eclairs	28-Feb-22	7,672	115	low			
27	Karlen McCaffrey	India	Fruit & Nut Bars	26-Jan-22	168	321	low			
28	Roddy Speechley	Canada	Spicy Special Slims	09-Feb-22	1,652	186	low			
29	Husein Augar	USA	Eclairs	07-Jul-22	4,025	112	low			
30	Dennison Crosswaite	New Zealand	White Choc	05-Jul-22	9,492	151	low			
31	Dennison Crosswaite	New Zealand	Manuka Honey Choco	15-Jun-22	5,061	301	low			
32	Van Tuxwell	New Zealand	85% Dark Bars	24-May-22	1,722	121	low			

Task 8: VLOOKUP & XLOOKUP

8a. Use VLOOKUP() to retrieve information

```
=VLOOKUP(C2, ProductCategory!$A2:$B24, 2, FALSE)
```

- VLOOKUP() searches for a value in the first column of a range and returns a value from another column in the same row.
- Here, it looks up the Product name (C2) in the ProductCategory table and returns its Category.
- The FALSE argument forces an exact match, which is important for text lookups like product names.
- VLOOKUP only searches to the right of the lookup column, which is one of its main limitations.

8b. Use XLOOKUP() to retrieve information

```
=XLOOKUP(C2, ProductCategory!A2:A24, ProductCategory!B2:B24)]
```

- XLOOKUP() is a modern replacement for VLOOKUP that searches a lookup array and returns a value from a matching return array.
- Here, it looks up the Product name (C2) in column A of ProductCategory and returns the matching Category from column B.

• Unlike VLOOKUP, XLOOKUP can look both left and right, and does not depend on column position/index numbers.

• XLOOKUP performs an exact match by default, so it does not require an extra argument like VLOOKUP's FALSE.

8c. Demonstrate the difference in usage

```
VLOOKUP: =VLOOKUP(C2, ProductCategory!$A2:$B24, 2, FALSE)
XLOOKUP: =XLOOKUP(C2, ProductCategory!A:A, ProductCategory!B:B)
```

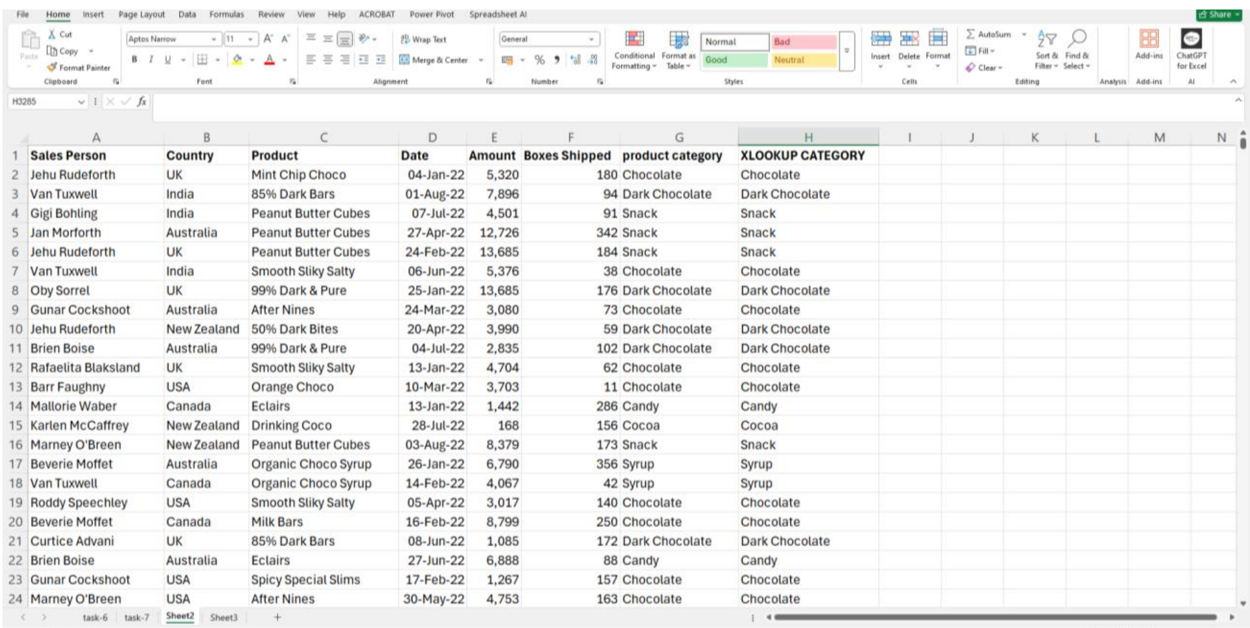
• VLOOKUP requires counting the column index number of the return column, which breaks if columns are inserted or removed.

• XLOOKUP references the return column directly, so it is more resilient to changes in table structure.

• VLOOKUP can only look up values to the right of the search column; XLOOKUP can look in either direction.

• Both return the same result here, but XLOOKUP is generally the more flexible, modern choice where available.

Task-8 Output:



Sales Person	Country	Product	Date	Amount	Boxes Shipped	product category	XLOOKUP CATEGORY
Jehu Rudeforth	UK	Mint Chip Choco	04-Jan-22	5,320	180	Chocolate	Chocolate
Van Tuxwell	India	85% Dark Bars	01-Aug-22	7,896	94	Dark Chocolate	Dark Chocolate
Gigi Bohling	India	Peanut Butter Cubes	07-Jul-22	4,501	91	Snack	Snack
Jan Morforth	Australia	Peanut Butter Cubes	27-Apr-22	12,726	342	Snack	Snack
Jehu Rudeforth	UK	Peanut Butter Cubes	24-Feb-22	13,685	184	Snack	Snack
Van Tuxwell	India	Smooth Sliky Salty	06-Jun-22	5,376	38	Chocolate	Chocolate
Oby Sorrel	UK	99% Dark & Pure	25-Jan-22	13,685	176	Dark Chocolate	Dark Chocolate
Gunar Cockshoot	Australia	After Nines	24-Mar-22	3,080	73	Chocolate	Chocolate
Jehu Rudeforth	New Zealand	50% Dark Bites	20-Apr-22	3,990	59	Dark Chocolate	Dark Chocolate
Brien Boise	Australia	99% Dark & Pure	04-Jul-22	2,835	102	Dark Chocolate	Dark Chocolate
Rafaelita Blaksland	UK	Smooth Sliky Salty	13-Jan-22	4,704	62	Chocolate	Chocolate
Barr Faughny	USA	Orange Choco	10-Mar-22	3,703	11	Chocolate	Chocolate
Mallorie Waber	Canada	Eclairs	13-Jan-22	1,442	286	Candy	Candy
Karlen McCaffrey	New Zealand	Drinking Coco	28-Jul-22	168	156	Cocoa	Cocoa
Marney O'Breen	New Zealand	Peanut Butter Cubes	03-Aug-22	8,379	173	Snack	Snack
Beverie Moffet	Australia	Organic Choco Syrup	26-Jan-22	6,790	356	Syrup	Syrup
Van Tuxwell	Canada	Organic Choco Syrup	14-Feb-22	4,067	42	Syrup	Syrup
Roddy Speechley	USA	Smooth Sliky Salty	05-Apr-22	3,017	140	Chocolate	Chocolate
Beverie Moffet	Canada	Milk Bars	16-Feb-22	8,799	250	Chocolate	Chocolate
Curtice Advani	UK	85% Dark Bars	08-Jun-22	1,085	172	Dark Chocolate	Dark Chocolate
Brien Boise	Australia	Eclairs	27-Jun-22	6,888	88	Candy	Candy
Gunar Cockshoot	USA	Spicy Special Slims	17-Feb-22	1,267	157	Chocolate	Chocolate
Marney O'Breen	USA	After Nines	30-May-22	4,753	163	Chocolate	Chocolate

Task 9: Pivot Table & Charts

9a. Create a Pivot Table and summarize the data

```
Insert -> PivotTable -> select data range -> New Worksheet

Rows: Country
Values: Sum of Amount
```

• A Pivot Table summarizes large datasets by automatically grouping and aggregating values.

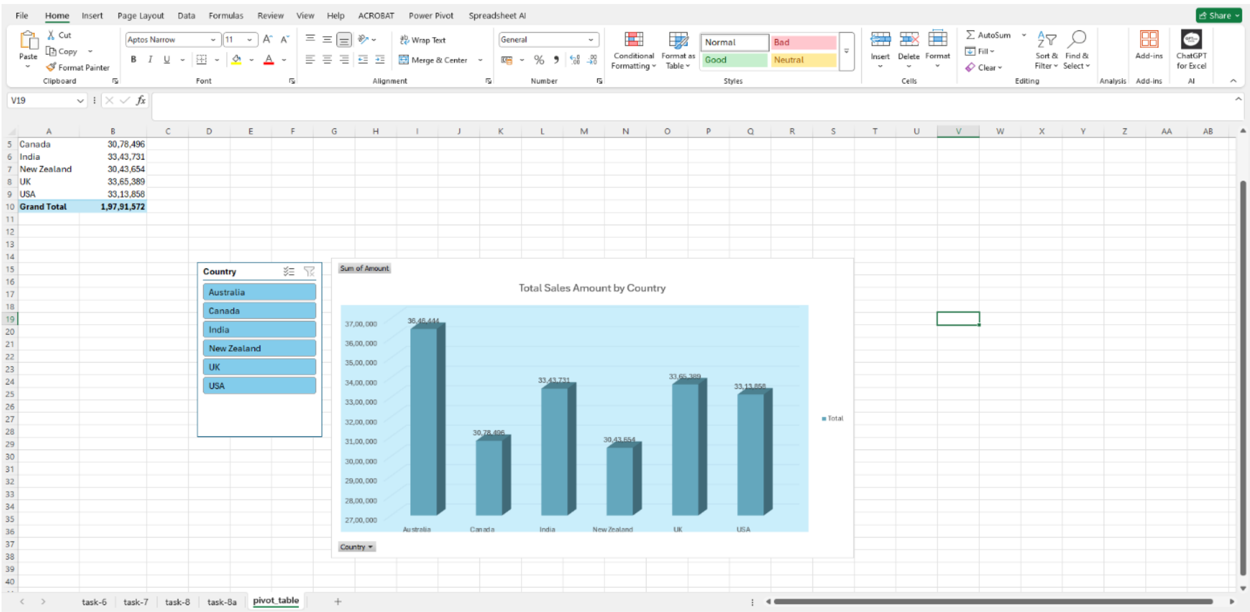
• Here, it totals the Amount column for each Country without writing any formulas manually.

- Fields can be dragged between Rows, Columns, Values, and Filters to reshape the summary instantly.
- Pivot Tables update automatically when the source data changes and the table is refreshed.

9b. Create a chart based on the analyzed data

```
Select PivotTable data -> Insert -> Recommended Charts -> Clustered Column Chart
```

Task-9 output:



- A chart turns the summarized Pivot Table data into a visual format that is easier to interpret.
- Here, a column chart compares total sales amount across different countries.
- Charts make it easier to quickly spot the highest and lowest performing categories.
- Because the chart is linked to the Pivot Table, it updates automatically when the data refreshes.